

AST 111 Lab Extra Credit

Assignment. Please read these instructions before proceeding! If you wish to receive optional extra credit **in the Lab portion of AST 111 only**, please complete these problems and send me your full solutions by **11:59pm on November 21st, 2025**. I will not accept late submissions. You must send me your solutions via email – **in the subject line you must include “AST 111 Lab Extra Credit Problems”**. These problems will test your understanding of the content we have covered this semester – you are encouraged to consult your lab scripts.

Expectation. Your final solutions must be typeset using Google Docs, Microsoft Word, L^AT_EX, or another typesetting language where you can properly type math. The expectation is that you provide complete solutions for each problem you choose to complete and clearly state any assumptions you make and/or formulas you apply. Your solutions should be complete (*i.e.*, responsive to every part of each problem) and **symbolic** until the end, at which point you may “plug in” numbers as needed. Failure to do these or properly typeset solutions will result that problem being excluded from considerations of extra credit.

Grading. You will be graded on the accuracy of your final answer for each problem and the completeness of the solutions used to get there. I should be able to follow your work smoothly without any questions of how you get from one place to another. You may change your letter grade by at most an entire letter grade by completing this assignment well. So, for example, if you have a C+ in the lab section and your work here is exceptional, you will receive a B+ in the lab section. You are not required to complete all problems for extra credit, but only those with solutions submitted will be considered.

Warning. As in the rest of the lab, **AI usage is expressly forbidden, but you are free to use any and all lab materials you have collected over the semester**. If there is suspicion of AI usage on any part of your solutions, you will receive no extra credit. This is done at my discretion, but rest assured that (a) if your solutions are done by you and done well this will not be a problem, and (b) there are objective answers to these problems. There is one and only one way to correctly work these problems, but you have the skills necessary to complete them if you have paid attention this semester. Everything you need to know has been discussed at some point this semester and you have all of the materials needed to work these. Please email me if there are issues!

... When I typed these, I looked through the lab scripts to make sure I only tested concepts we would’ve encountered during the semester. That being said, these are not meant to be problems that are recasts of problems you have already done. They are the same concepts you are familiar with from the semester, but in an applied setting; *i.e.*, we talked about Kepler’s Laws and what they look like, but now you are going to derive them. You have to work for extra credit!

Good luck!

Problems.

1. The kinetic energy of an object of mass m moving at speed v is given by

$$T = \frac{1}{2}mv^2. \quad (1)$$

The gravitational potential energy for an object of mass m at height h is given by

$$U = mgh \quad (2)$$

where g is the acceleration due to gravity.

- If energy is conserved, derive an expression for the speed of an impactor when it hits the surface of a planetary body.
- What would happen to this impact velocity if the mass of the impactor was tripled? What about if its height was tripled?
- Measurements find that

$$\log_b d = k^{-1} \log_b T \quad (3)$$

describes the crater diameter d empirically. Write Equation (3) as a power law of the form $d = T^\alpha$ where α is a constant. Does impactor mass or impactor velocity affect the crater diameter more?

2. You are given unlimited funding from Congress to design a state-of-the-art space telescope! You want to build it so that you can see small, faint objects in high resolution. The first decisions to be made are the basic specifications it will be built around.
- The previous flagship space telescopes, the *Hubble Space Telescope (HST)* and the *James Webb Space Telescope (JWST)*, have primary mirror diameters of 2.4 meters and 6.5 meters, respectively. Approximately what is the ratio of *JWST*'s light-gathering power to *HST*'s light gathering power? Should you choose a larger or smaller primary mirror to see fainter objects?
 - You are interested in observing a distant galaxy cluster. It is at a distance of 4.3 billion light years from Earth, around which the telescope will orbit. If it is 3.5 arcminutes on a side on the sky, what are its linear dimensions in light years? It may be useful to know that

$$\sin \theta \sim \theta \text{ for small } \theta. \quad (4)$$

Hint: right triangle trigonometry will be useful here!

- You want to make a beautiful press release image of the data you get of this galaxy cluster, so you have to make sure the field of view is right for imaging. If the telescope has a constant focal length, should you design it to have an "eyepiece focal length" for its camera that is large or small? Why?
Hint: the cluster is small on the sky, so what does this say about the field of view you want?
3. You are observing a quasar, the accreting supermassive black hole at the center of an active galaxy. Quasars have hot, dense regions of dust around them that absorb and re-radiate heat energy from the black hole. In order to make statements about its properties, you have resorted to spectroscopy.

- You observe a prominent $H\alpha$ line in your quasar spectra, which has $n = 3 \rightarrow 2$. What is the change in the electron's energy if

$$E_n = -\frac{\mathcal{R}}{n^2}, \quad \mathcal{R} = 13.6 \text{ eV} \quad (5)$$

is the energy of an electron with quantum number n ? What about the emitted photon's wavelength?

- If there is a foreground cloud of gas in your line of sight, what kind of spectrum should you expect to see? Explain your reasoning.
- You measure the temperature of the quasar to be 350 million kelvin. What wavelength does its emission peak, and where does this peak fall on the electromagnetic spectrum (radio, ultraviolet, microwave, or somewhere else – give a specific region)? How many times hotter is this quasar than the surface of the sun, whose emission peaks at 500 nm? *Hint: Wien's displacement law makes this problem a breeze.*

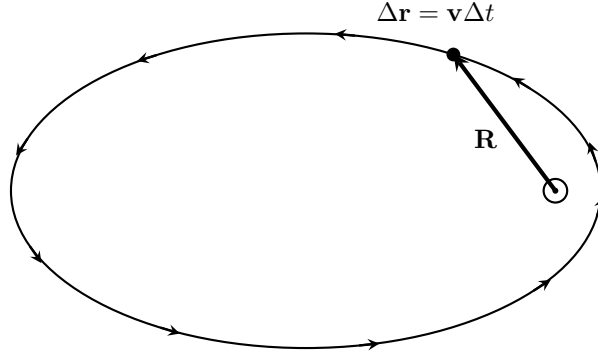
4. You are aware of Kepler's 3 Laws. Now you will derive them.

- a. For gravity, an attractive inverse-square law force, the radial distance r from a focus as a function of angle φ is given by

$$r(\varphi) = \frac{c}{1 + \epsilon \cos \varphi} \quad (6)$$

for a constant $c > 0$. The number ϵ is the orbit's eccentricity. Choose some arbitrary value for a and graphically show that, for $\epsilon < 1$, this is the equation of an ellipse. This is Kepler's 1st Law: all objects in bound orbits move in ellipses with the more massive object at a focus.

- b. Imagine a planet orbiting its star in an elliptical orbit with radius $\mathbf{r}$ and change in position $\Delta \mathbf{r} = \mathbf{v} \Delta t$ as shown in the Figure.



Given that the planet moves and sweeps out an area

$$\Delta A = \frac{1}{2} |\Delta \mathbf{r} \times \mathbf{r}| = \frac{1}{2} |\mathbf{v} \Delta t \times \mathbf{r}| \quad (7)$$

and that

$$\ell = \mu |\mathbf{r} \times \mathbf{v}|, \quad (8)$$

show that

$$\frac{\Delta A}{\Delta t} = \frac{\ell}{2\mu}. \quad (9)$$

This is Kepler's 2nd Law: "equal areas in equal times". *Hint: multiply Equation (7) by Δt ; the $\times$ symbol works as ordinary multiplication inside the $||$ for our purposes.*

- c. For the period of an orbiting body τ , we can write that

$$\frac{\Delta A}{\Delta t} = \frac{\pi ab}{\tau} \quad (10)$$

where πab is the area of an ellipse. Using that

$$a = \frac{c}{1 - \epsilon^2}, \quad b = \frac{c}{\sqrt{1 - \epsilon^2}}, \quad c = \frac{\ell^2}{\gamma \mu} \quad (11)$$

for

$$\gamma = GMm, \quad \mu = \frac{Mm}{M + m}, \quad (12)$$

show that

$$\tau^2 = \frac{4\pi^2}{G(M + m)} a^3. \quad (13)$$

This is Kepler's 3rd Law: the square of the period is proportional to the cube of the semi-major axis length. *Hint: this is just some substitutions and some algebra! It doesn't bite.*

5. The flux F of an object (you can think of this as being the “amount of light” you see coming from it) is given by

$$F = \frac{L}{4\pi d^2} \quad (14)$$

where L is the energy (light) output of the object (it is called a luminosity) and d is its distance from you.

- Show that the units of flux are that of $[\text{energy}] \cdot [\text{area}]^{-1} \cdot [\text{time}]^{-1}$. Explain why the flux of a closer object is greater than that of a further one for a given luminosity. Conceptually, why is there a factor of 4π in the denominator?
- The Large Magellanic Cloud (LMC), one of the Milky Way’s satellite galaxies about 160,000 light years away, is an active “stellar nursery” with a high star-formation rate. Suppose 2 stars in an LMC star cluster are separated by half a lightyear. Can *HST*, an optical space telescope, resolve these stars as distinct objects? *Hint: use the mirror diameter given in a previous problem.*
- When looking at things outside of the Milky Way, we have to contend with foreground *optical dust extinction* A_V , where dust between us and the object we are viewing diminishes the flux we receive from it. Suppose for the LMC that this is given by a diminutive multiplicative factor $\mathcal{D}$ on the flux given by

$$\mathcal{D} = 10^{-0.4A_V} \quad (15)$$

for a given A_V . If $A_V = 0.3$, how much cooler would a star in the LMC appear if dust extinction is not corrected for?